Network Science

Lecture 09: Node Representations and Graph Embeddings

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From Structure to Representation

Every previous lecture analysed the graph as a **combinatorial object**: edges, triangles, communities, cuts, shortest paths. Today we ask the dual question:

> **Can every node be represented as a *point* in $\mathbb{R}^d$ such that graph structure is preserved as geometry?**
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> If two nodes are "similar" in the graph (same community, similar role, co-visited by walks), they should be *close* as vectors.

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This is **graph representation learning** — turning a discrete object into a continuous one so that we can use every tool from classical ML: clustering, classification, retrieval, generative models.

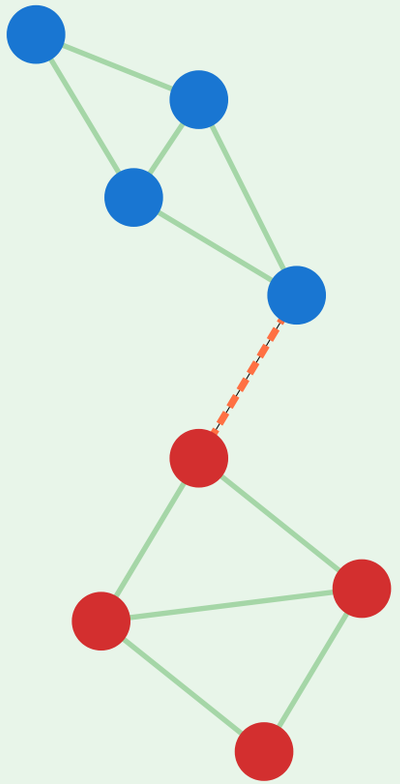
The Question — Made Visual

Laplacian Eigenmaps — From Graph to Geometry

Eigenvectors of the Laplacian embed each node as a point in $\mathbb{R}^d$

1. Graph G

two weakly connected clusters



colour = sign of Fiedler vector

2. Laplacian L

$L = D - A$

$$\begin{bmatrix} 2 & -1 & 0 & 0 & \cdots \\ -1 & 3 & -1 & -1 & \cdots \\ 0 & -1 & 3 & -1 & \cdots \\ 0 & -1 & -1 & 2 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

eigendecomposition

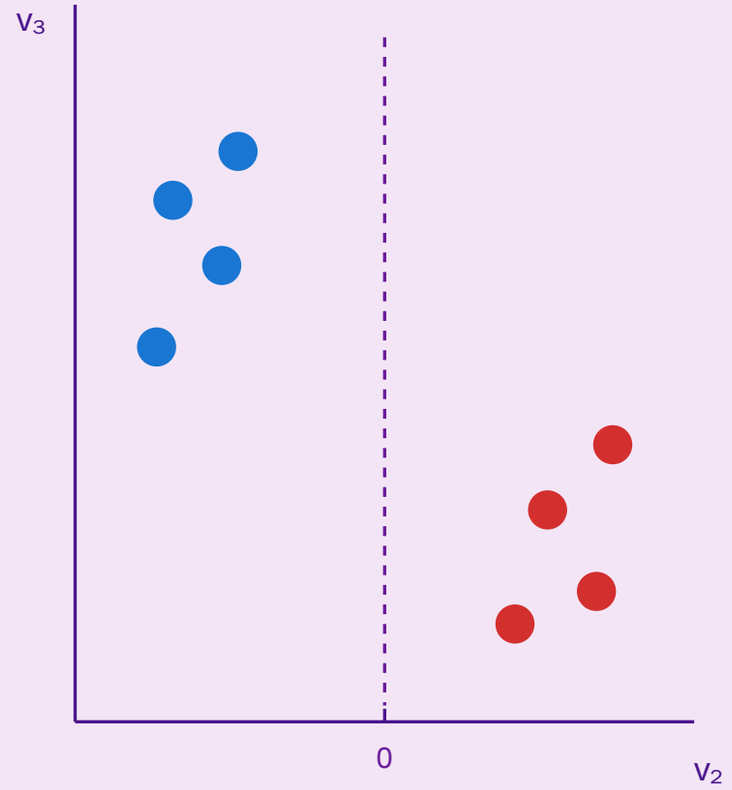
$L v_k = \lambda_k v_k$

$0 = \lambda_1 \le \lambda_2 \le \lambda_3 \le \dots$

take v_2, v_3 as coords

3. Embedding $\mathbb{R}^2$

$z_v = (v_2(v), v_3(v))$

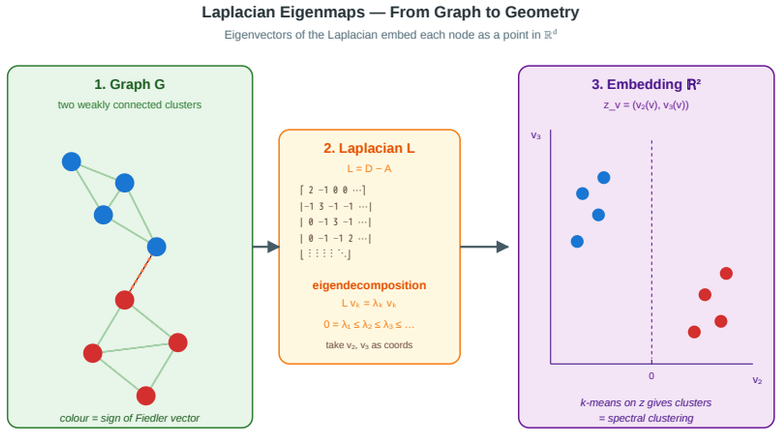


k-means on z gives clusters
= spectral clustering

A Sixth Gap — The Representation Gap

L07 (navigability) and L09 (representation) are the two faces of the same problem: making graphs *work* with local, vector-based operations.

Lecture	Gap	Strategy
L07	Navigational	Multi-scale links
L09	**Representation**	**Embed nodes in $\mathbb{R}^d$**
L08	Process-structure	Compare processes



Takeaway

Graph representation learning turns a discrete object into a continuous one. The goal is a map $z : V \to \mathbb{R}^d$ such that graph similarity $\approx$ vector similarity. Different methods capture different notions of "similar."

Spectral Foundations

Guiding Question: Can the eigenvectors of a graph matrix give us a natural geometry for its nodes?

The Graph Laplacian — One More Time

For an undirected graph $G = (V, E)$ with adjacency A and degree diagonal D:

$$L = D - A \qquad \text{(combinatorial Laplacian)}$$

$$L_{\text{sym}} = I - D^{-1/2} A D^{-1/2} \qquad \text{(normalised Laplacian)}$$

L is symmetric positive semi-definite. Its eigenvalues satisfy $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$.

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The eigenvectors corresponding to the smallest non-zero eigenvalues are the most informative: they encode the graph's *large-scale* structure.

Laplacian Eigenmaps (Belkin & Niyogi 2003)

Let $v_2, v_3, \ldots, v_{d+1}$ be the eigenvectors of L for the d smallest non-zero eigenvalues. Define

$$z_v = \big(v_2(v), v_3(v), \ldots, v_{d+1}(v)\big) \in \mathbb{R}^d$$

Each node gets a d-dimensional embedding.

Why this works. The embedding is the solution to

$$\min_{Z} \sum_{(u,v)\in E} |z_u - z_v|^2 \quad \text{s.t. } Z^\top D Z = I$$

— connected nodes end up close, and the constraint prevents the trivial solution $z = 0$.

Spectral Clustering = k-means on Laplacian Eigenmaps

Laplacian eigenmaps + k-means = **spectral clustering**.

- Nodes in the same community have similar low-frequency eigenvector values.
- k-means on $Z \in \mathbb{R}^{|V| \times d}$ recovers communities.
- This is how NCut (Shi & Malik 2000) and most image-segmentation algorithms work.

> **Spectral gap.** The jump $\lambda_{k+1} - \lambda_k$ indicates the "natural" number of communities. A large gap means the graph has *exactly* k clear clusters; a small gap means the partition is ambiguous.

Signed Laplacian — Bridge to L06

For **signed networks** (L06, structural balance), the *signed Laplacian* $\bar{L} = \bar{D} - A$ (with $\bar{d}_i = \sum_j |A_{ij}|$) encodes enmity as well as friendship.

Low eigenvectors of $\bar{L}$ embed nodes such that friends cluster and enemies separate — the continuous analogue of approximate structural balance.

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Kunegis et al. (2010) used this to detect communities in signed online networks (Slashdot, Epinions) where positive and negative edges coexist.

The Spectral Limitations

Why can't we just use spectral embeddings for everything?

1. **Cost.** Computing eigenvectors of a $10^9 \times 10^9$ Laplacian is infeasible. Best sparse solvers still take $O(|E| \cdot d)$ per iteration.
2. **Transductive.** A new node requires recomputing the whole spectrum — there is no "inference" step.
3. **No features.** Spectral embeddings use only graph structure; they ignore node attributes (text, images, tabular).
4. **Global.** All nodes see the same basis; local neighbourhood-specific patterns get averaged away.

Takeaway

Spectral methods embed each node using the graph Laplacian's low eigenvectors. They are principled, globally optimal, and reduce clustering to linear algebra — but they scale poorly, are transductive, and ignore node features.

Quick Check

You have a graph with 3 disconnected components of similar size.

1. What are the first three eigenvalues of the Laplacian L?
2. What do the first three eigenvectors tell you?
3. What does "spectral gap" mean for this graph?

Quick Check — Solution

$\lambda_1 = \lambda_2 = \lambda_3 = 0$ — the multiplicity of $\lambda = 0$ equals the number of connected components.

They span the space of piecewise-constant vectors (constant on each component). Any indicator vector of one component is a linear combination of them.

The gap is between $\lambda_3 = 0$ and λ_4. Large $\lambda_4 \rightarrow$ components are well-separated and internally well-knit. Small $\lambda_4 \rightarrow$ one component is close to splitting further.

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Random Walks as Representations — DeepWalk

Guiding Question: Can we use *samples* of graph structure (random walks) instead of spectral factorisation?

The Analogy — Graph ≈ Corpus

Language (word2vec)	Graph (DeepWalk)
Vocabulary	Node set V
Corpus	Collection of random walks
Sentence	One walk $v_1 v_2 \dots v_L$
Word	Node v_i
Context window	Walk window of size c
Skip-gram objective	Skip-gram objective

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Insight. If we can generate many random walks from each node, we can reuse the entire word2vec machinery to learn node embeddings.

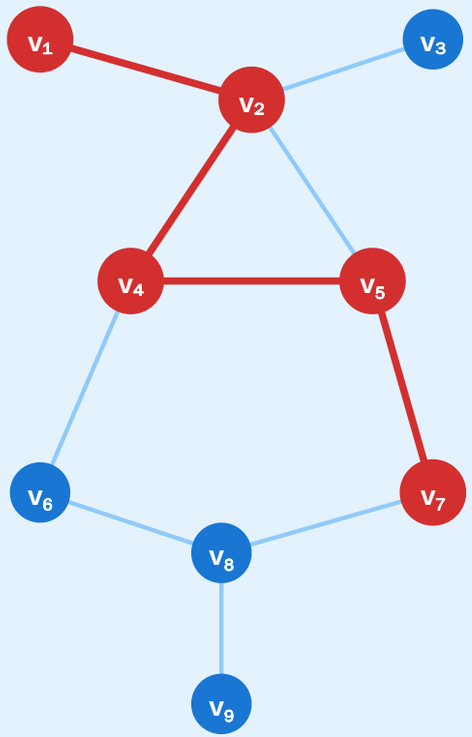
DeepWalk Pipeline

DeepWalk — Random Walks as "Sentences" for Skip-Gram

node ≈ word · walk ≈ sentence · co-occurrence → embedding

1. Random walks

length L from each node



walk: $v_1\ v_2\ v_4\ v_5\ v_7$

repeat y times per node

2. Skip-gram window

context c = 2 around centre

centre v_4 · context $\{v_1, v_2, v_5, v_7\}$

training pairs (v, context)

(v_4, v_1) (v_4, v_2)

(v_4, v_5) (v_4, v_7)

Skip-gram objective

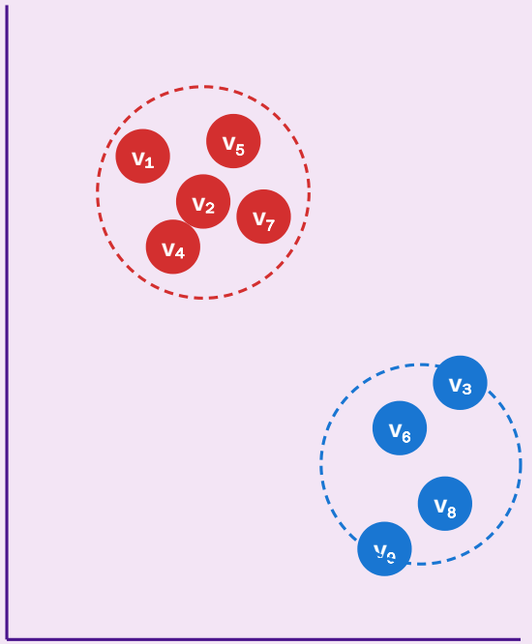
$$\max \Sigma \log P(u \mid v)$$

$$P(u|v) = \text{softmax}(z_u \cdot z_v)$$

"co-walking nodes have similar z"

3. Embeddings

$z_v \in \mathbb{R}^d$, $d \ll |V|$



co-walked nodes end up close in the embedding space

Perozzi et al. 2014

The Skip-Gram Objective

Given walks W_i and window c, learn vectors $z_v \in \mathbb{R}^d$ maximising

$$\mathcal{L} = \sum_i \sum_t \sum_{-c \le j \le c, j \neq 0} \log P(v_{t+j} \mid v_t)$$

with

$$P(u \mid v) = \frac{\exp(z_u \cdot z_v)}{\sum_{u' \in V} \exp(z_{u'} \cdot z_v)}$$

Dense softmax over $|V|$ is too expensive → **negative sampling** approximates it by k random non-context nodes per pair.

What DeepWalk Captures

Proposition (Qiu et al. 2018). DeepWalk with long walks is *implicitly factorising* the log of a shifted PPMI matrix built from the random-walk co-occurrence matrix.
This connects DeepWalk back to spectral methods: both minimise a matrix-factorisation objective — the matrices just differ.

Key result. Random-walk methods and spectral methods sit on the same theoretical spectrum. DeepWalk is an approximate, sample-based spectral embedding of the walk co-occurrence matrix.

Takeaway

DeepWalk treats a random walk as a sentence and reuses word2vec's skip-gram. It is scalable (SGD, no eigendecomposition) and unsupervised. Under the hood, it is factorising a co-occurrence matrix — the same machinery as spectral embedding, in a sampled form.

node2vec — Biased Walks

Guiding Question: Should "similar" mean *same community* or *same structural role*? Can one embedding capture both?

Two Notions of Similarity

Homophily (community). Two nodes are similar if they are in the same densely connected region — even if structurally different (one is a hub, the other is a leaf).
DFS-like walks wander far into one community.

Structural equivalence (role). Two nodes are similar if they play the *same role* — both are hubs, both are bridges — even if in different communities.
BFS-like walks stay local and sample neighbourhoods.

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DeepWalk's uniform walks give a blur between the two. node2vec lets us choose.

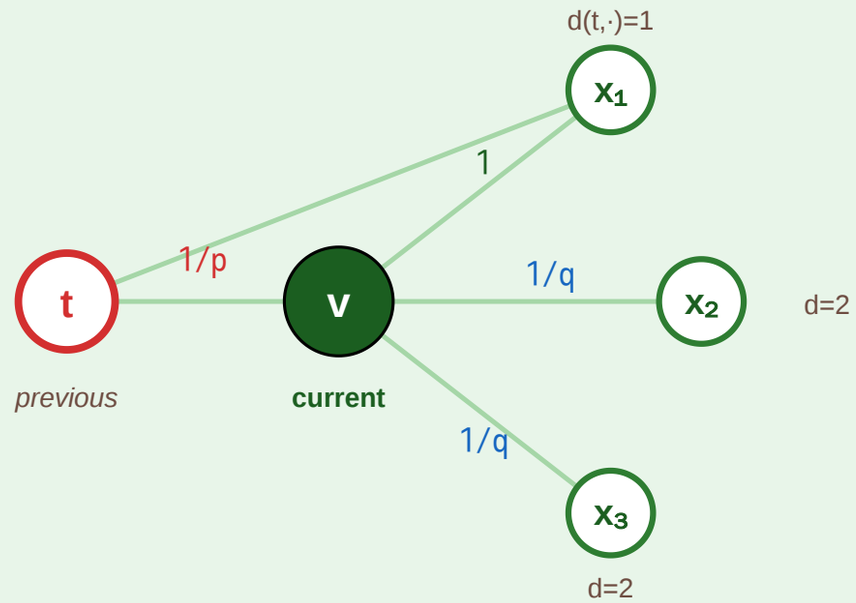
The Biased Walk

node2vec — Biased Random Walks with Return (p) and In-Out (q)

tune the walk between BFS-like (structural) and DFS-like (community) exploration

1. Second-order transition

biased by where we came from (t)



α(t → v → ·) based on distance d(t,·):

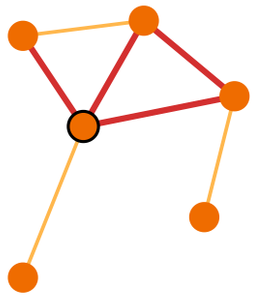
`d=0 (back to t): 1/p · d=1: 1 · d=2: 1/q`

2. Two regimes, two embeddings

p, q shape what "similar" means

small q → BFS-like

stay local, explore neighbours



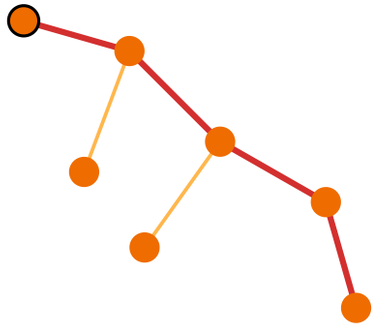
captures structural roles

hubs, bridges → similar z

"homophily by role"

small p → DFS-like

wander far, reveal communities



captures communities

same cluster → similar z

Grover & Leskovec 2016 — one knob, two inductive biases

node2vec in Production

Despite its age, node2vec is still widely deployed because:

- **Scalable.** Runs on billions of edges with sampled walks + SGD.
- **No features required.** Works on raw adjacency, making it a sensible *baseline* for any graph task.
- **Interpretable knobs.** p and q correspond to clear exploration policies.

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> **Limitation.** Like DeepWalk, node2vec is **transductive** and **feature-free**. A new node requires a new walk + retraining; node attributes (text, numeric features) are ignored. These shortcomings motivate GNNs.

Graph Neural Networks — Message Passing

Guiding Question: Can we learn node embeddings that *also* use node features — end-to-end with any downstream task?

The Message-Passing Framework

At each layer l, every node v updates its representation by aggregating messages from its neighbours $\mathcal{N}(v)$:

$$h_v^{(l+1)} = \mathrm{UPDATE}\Big(h_v^{(l)},; \mathrm{AGG}\big(h_u^{(l)} : u \in \mathcal{N}(v)\big)\Big)$$

- AGG is a permutation-invariant function: sum, mean, max, or attention.
- UPDATE is usually a learnable linear transform + nonlinearity: $\sigma(W[h_v, m_v])$.
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Stacking L layers gives each node a *receptive field* of L hops — its final embedding reflects its L-hop neighbourhood.

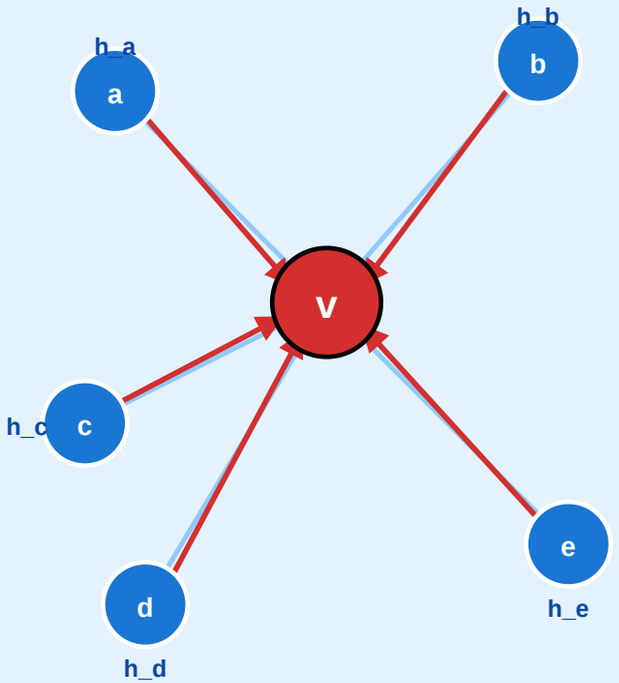
Message Passing — Visual

Graph Neural Networks — Message Passing

$h^{(l+1)}_v = \text{UPDATE}(h^{(l)}_v, \text{AGG}(\{ h^{(l)}_u : u \in N(v) \}))$

1. Neighbourhood of v

each node has a feature vector h_v

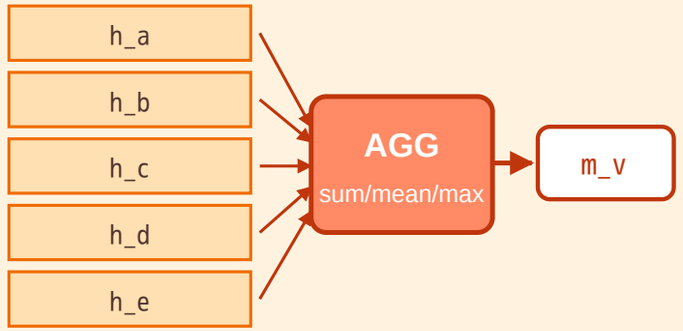


messages travel along edges

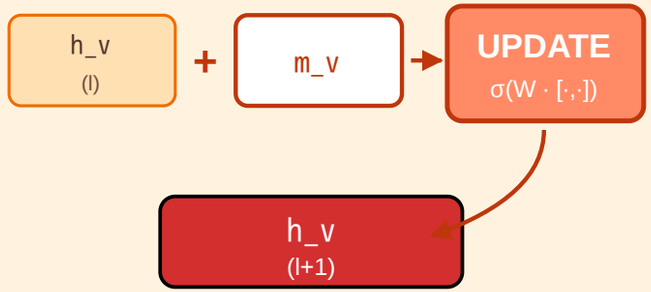
N(v) = { a, b, c, d, e }

2. Aggregate · Update

permutation-invariant pooling



then combine with own state



one GNN layer = one hop further

3. Stack L layers

receptive field = L hops

Layer 1 · 1-hop

Layer 2 · 2-hop

Layer 3 · 3-hop

Common variants

- GCN — normalised sum
- GraphSAGE — sampling
- GAT — attention weights
- GIN — sum + MLP

too many layers →
over-smoothing
all nodes collapse to same z

Four GNN Variants in One Line Each

- **GCN** (Kipf & Welling 2017) — AGG = normalised sum: $\tilde{A} = D^{-1/2}(A + I)D^{-1/2}$; UPDATE = $\sigma(\tilde{A}HW)$.
- **GraphSAGE** (Hamilton et al. 2017) — AGG on a *sampled* subset of neighbours; scales to very large graphs; inductive by design.
- **GAT** (Veličković et al. 2018) — AGG weighted by learned *attention* α_{uv}.
- **GIN** (Xu et al. 2019) — AGG = sum; UPDATE = MLP. Provably as powerful as the Weisfeiler-Lehman graph isomorphism test.

The Over-Smoothing Problem

Stacking too many GNN layers kills the embedding.

- After L layers, each node's embedding averages over its L-hop neighbourhood.
- For random graphs, all nodes eventually see the entire graph → all embeddings collapse to the same vector.
- This happens already at $L = 3$–5 for typical benchmarks.

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Remedies. Residual connections (JKNet), edge-weight normalisation, explicit node-pair features, or simply using fewer layers (1–3 is often enough).

Takeaway

GNNs learn node embeddings by repeated message passing — each layer extends the receptive field by one hop. They are inductive, feature-aware, and end-to-end trainable. But they suffer from over-smoothing: more layers is not always better.

Quick Check

You train a 2-layer GCN on a citation network to predict paper topics.

1. What is the receptive field of each paper's final embedding?
2. Why would stacking 10 layers hurt performance?
3. If you add a new paper tomorrow, can you predict its topic without retraining?

Closing the Loop — Embeddings Meet HNSW

Guiding Question: How do the embeddings we just built power modern retrieval systems — and what does this have to do with Kleinberg's grid?

The Full Pipeline

1. **Build embeddings** — spectral, DeepWalk, node2vec, or a trained GNN produces $z_v \in \mathbb{R}^d$ for each node.
2. **Build an index** — HNSW (L07) or similar ANN structure over $z_v : v \in V$.
3. **Query** — encode query q as z_q, return top-k nearest z_v in $O(\log n)$.
4. **Use** — link prediction, recommendation, RAG retrieval, semantic search.

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The same pipeline powers **product recommendation** (Alibaba, Amazon), **friend suggestion** (LinkedIn, Meta), **fraud detection** (PayPal), and **retrieval-augmented LLMs** (2024–2026 production stacks).

Why Geometry Matters

HNSW needs the embedding space to have the **right geometry** for greedy descent to work:

- **Similar nodes close, dissimilar nodes far** — true by construction in spectral / DeepWalk / GNN embeddings.
- **Low intrinsic dimension** — if embeddings live on a low-dimensional manifold inside $\mathbb{R}^d$, greedy search is efficient. Otherwise we hit the curse of dimensionality.
- **Smoothness** — continuous similarity (not sharp drops), which is why normalisation, temperature, and contrastive losses matter.

The *quality* of the embedding determines whether the index is actually navigable.

Takeaway

Embeddings (L09) and navigable indexes (L07) are the two halves of the modern retrieval pipeline. Spectral, DeepWalk, node2vec, and GNN methods produce the vectors; HNSW descends them. Kleinberg's insight ($r = d$) reappears as an engineering guarantee in HNSW's layer structure.

Wrap-Up — The Representation Arc

Method	Uses	Scales to	Inductive	Feature-aware
Laplacian Eigenmaps	eigendecomposition	$\leq 10^6$ nodes	✗	✗
DeepWalk / node2vec	random walks + skip-gram	10^9 edges	✗	✗
GCN / GraphSAGE / GAT	message passing	10^9 edges	✓	✓
Combined w/ HNSW	any of the above → ANN index	10^9 vectors	✓	✓

Each row *adds* a capability without replacing the previous row — classical spectral embeddings remain the theoretical benchmark, random-walk methods remain the unsupervised baseline, GNNs add features, HNSW adds retrieval.

Looking Ahead: Exercise and Beyond

The exercise for this lecture compares Laplacian Eigenmaps, node2vec, and a small GCN on a shared dataset. Real graph-ML research in 2024–2026 has moved on to **graph transformers**, **graph foundation models**, and **heterogeneous GNNs** — but all rest on the message-passing core you now understand.

Reading

Chapter 23 of Hamilton's *Graph Representation Learning* (2020) covers spectral, random-walk, and GNN embeddings. Perozzi et al. (2014) is DeepWalk. Grover & Leskovec (2016) is node2vec. Kipf & Welling (2017) introduces GCN. Hamilton et al. (2017) introduces GraphSAGE. Malkov & Yashunin (2018) is the HNSW paper that closes the loop to L07.